Forecasting with stationary models I

This version: October 29, 2026

ECMA 31150, Spring 2026

University of Chicago

Department of Economics

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Plan for this slide deck

1. What is time series analysis?
2. Definition of time series
3. Stationarity
4. Serial correlation, autocorrelation function

What is time series analysis?

- Time series: data with *time ordering*.
- Observations often not i.i.d. or even uncorrelated over time. "Random sampling" from super-population not relevant.
- Topics that receive more emphasis in time series analysis than in cross-sectional econometrics:
  - Forecasting (prediction).
  - Small sample sizes.
  - Strong dependence over time.
  - Unstable models over time.
  - Data always observational. Must look hard for "natural experiments".
  - Theory (structural models) to guide/complement causal inference.

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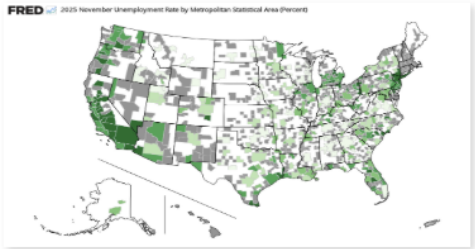
At a Glance



Consumer Price Index for All Urban Consumers: All Items



The Implementation of Monetary Policy



Unemployment Rate by Metro Area

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Real Gross Domestic Product (GDPC1)

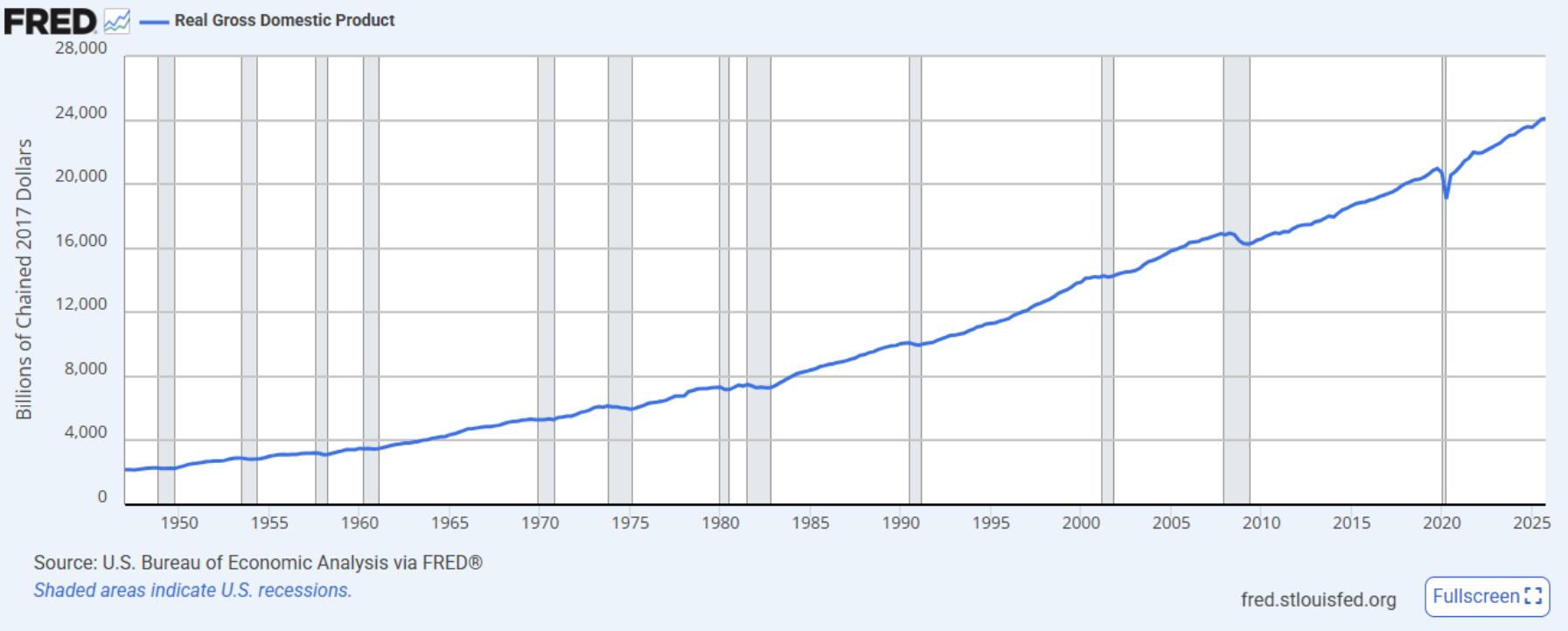
Observations
Q4 2025: **24,111.830**
Updated: Feb 20, 2026 7:48 AM CST
Next Release Date: Mar 13, 2026

Units:
Billions of Chained 2017 Dollars,
Seasonally Adjusted Annual Rate

Frequency:
Quarterly

1Y | 5Y | 10Y | Max
1947-01-01 to 2025-10-01

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Shaded areas indicate U.S. recessions.

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Real Gross Domestic Product (GDPC1)

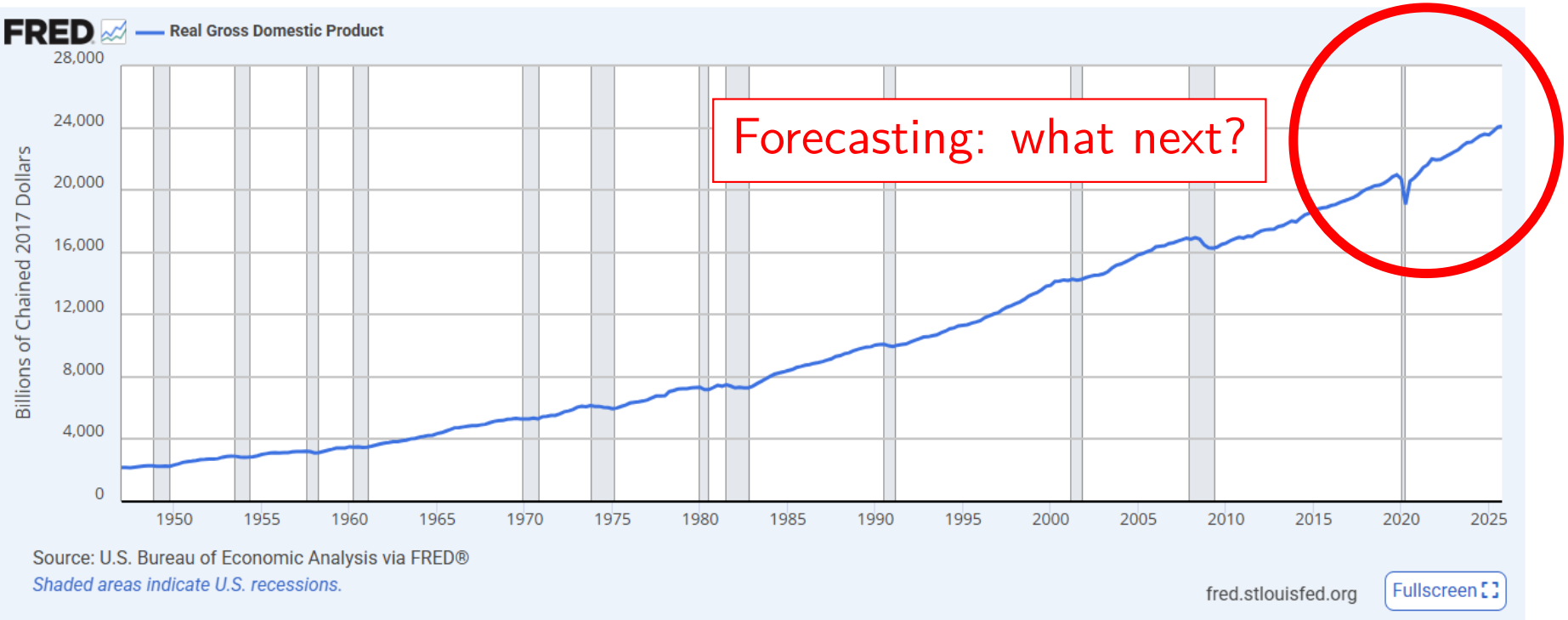
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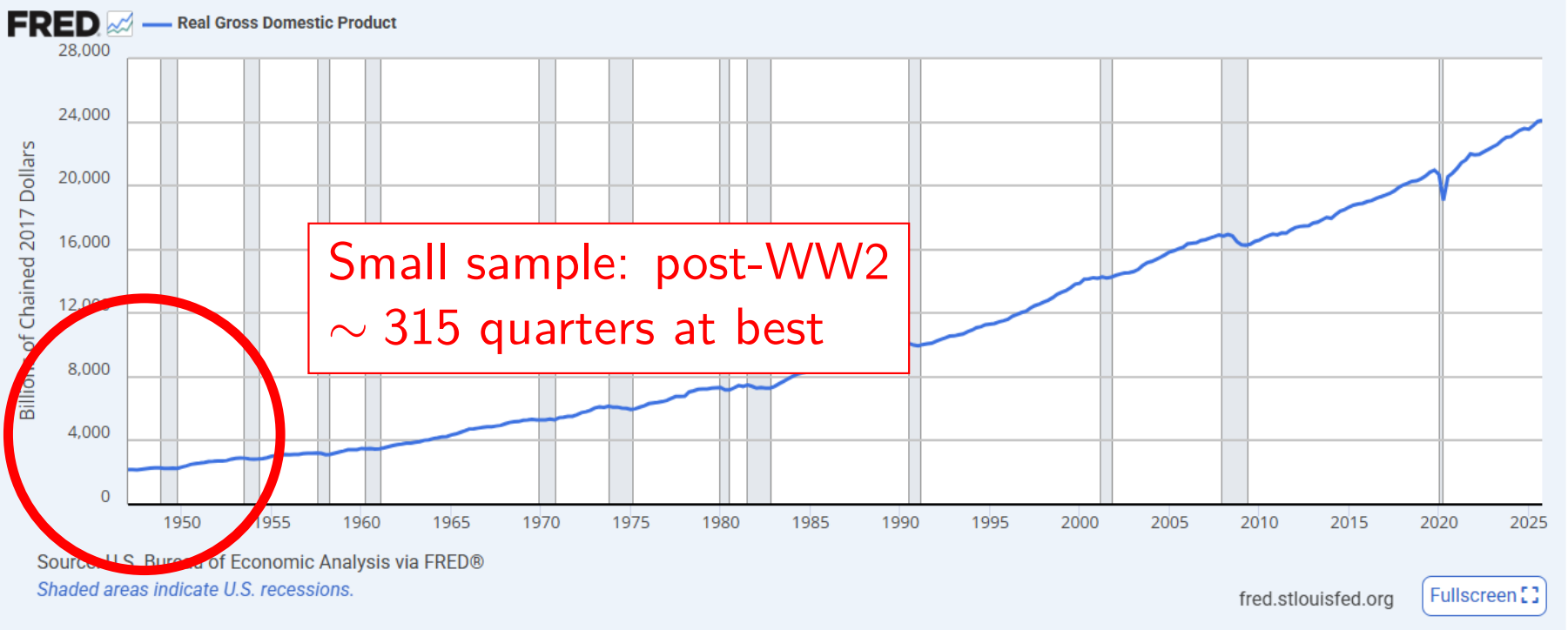
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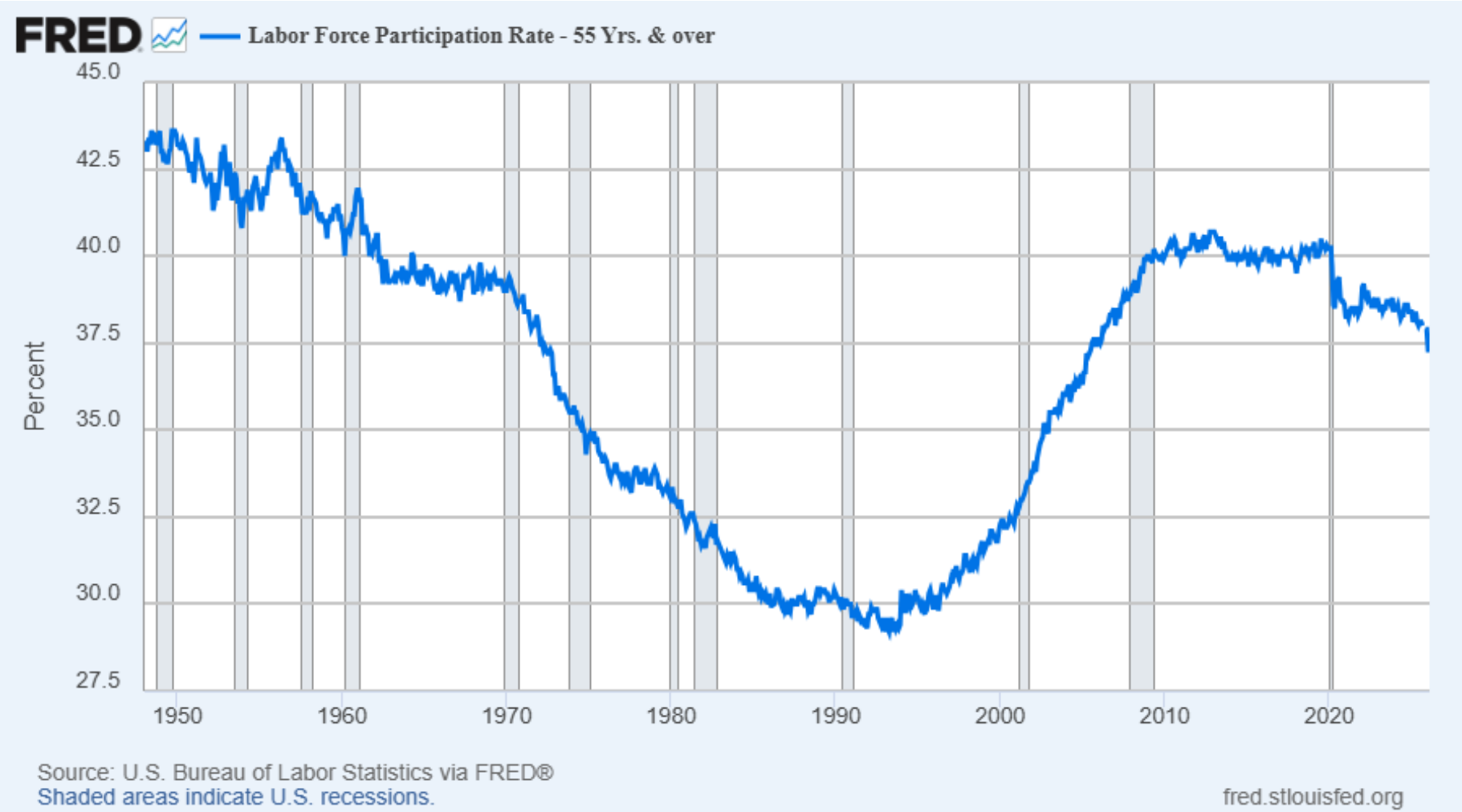
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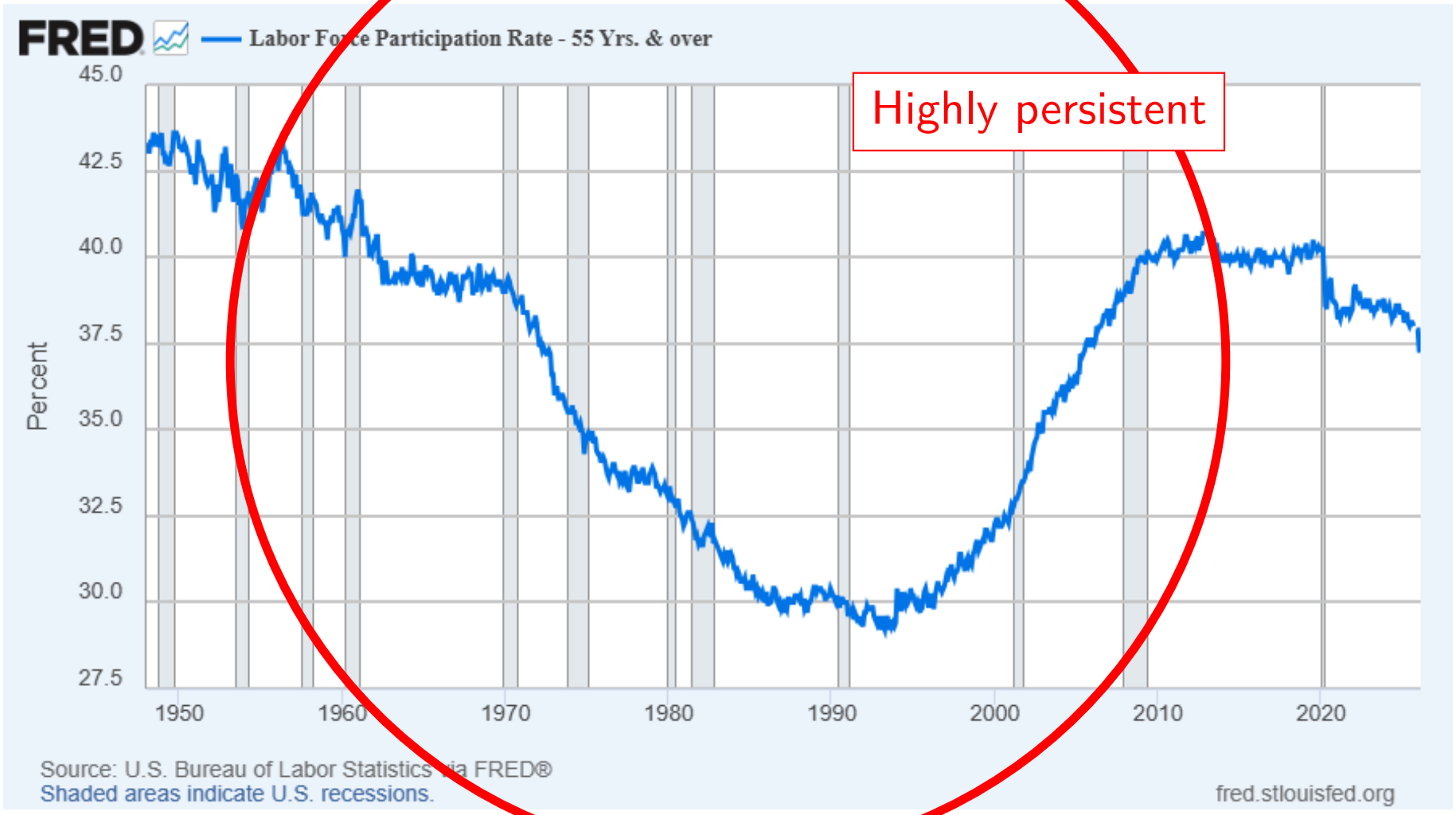
Frequency:
Quarterly

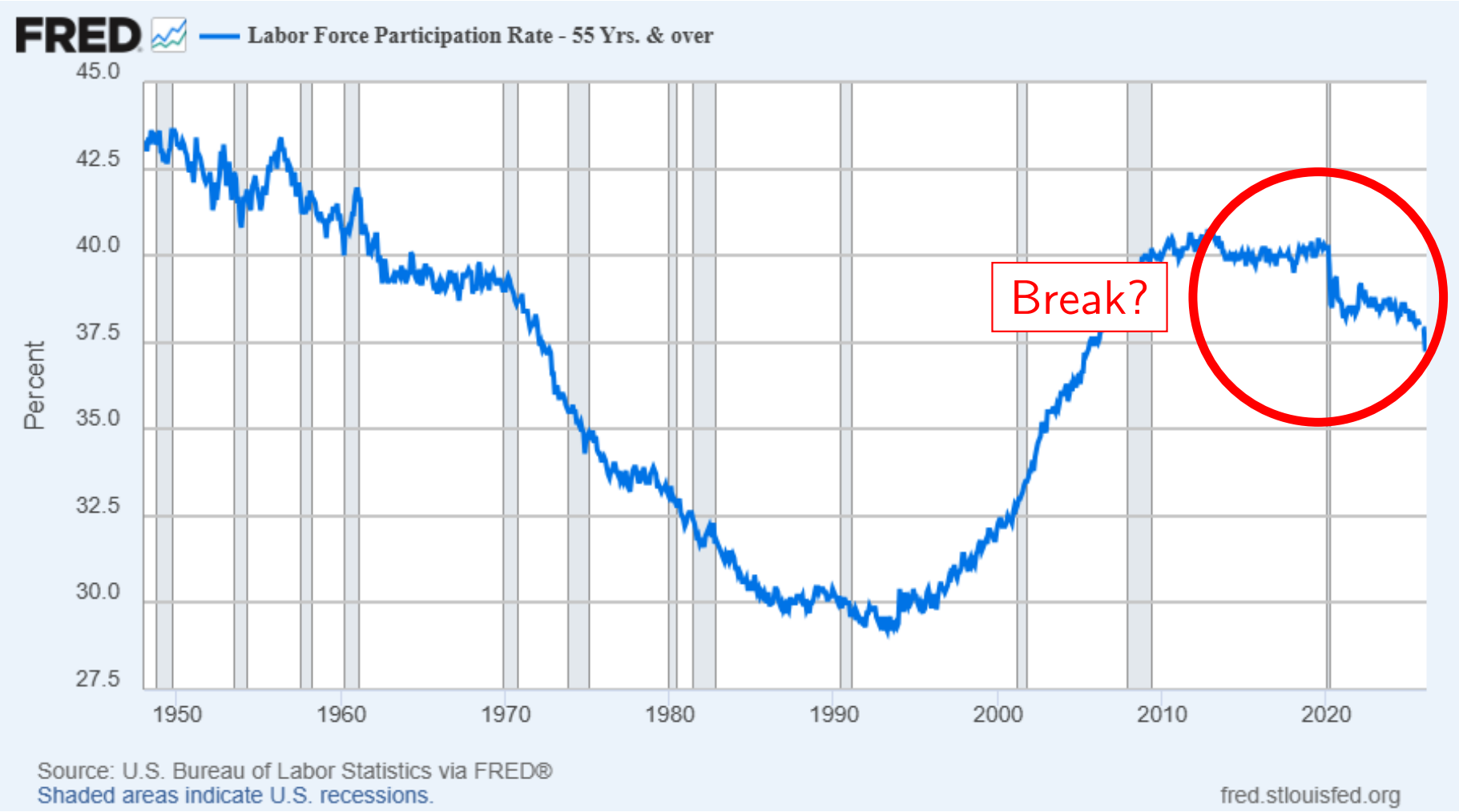
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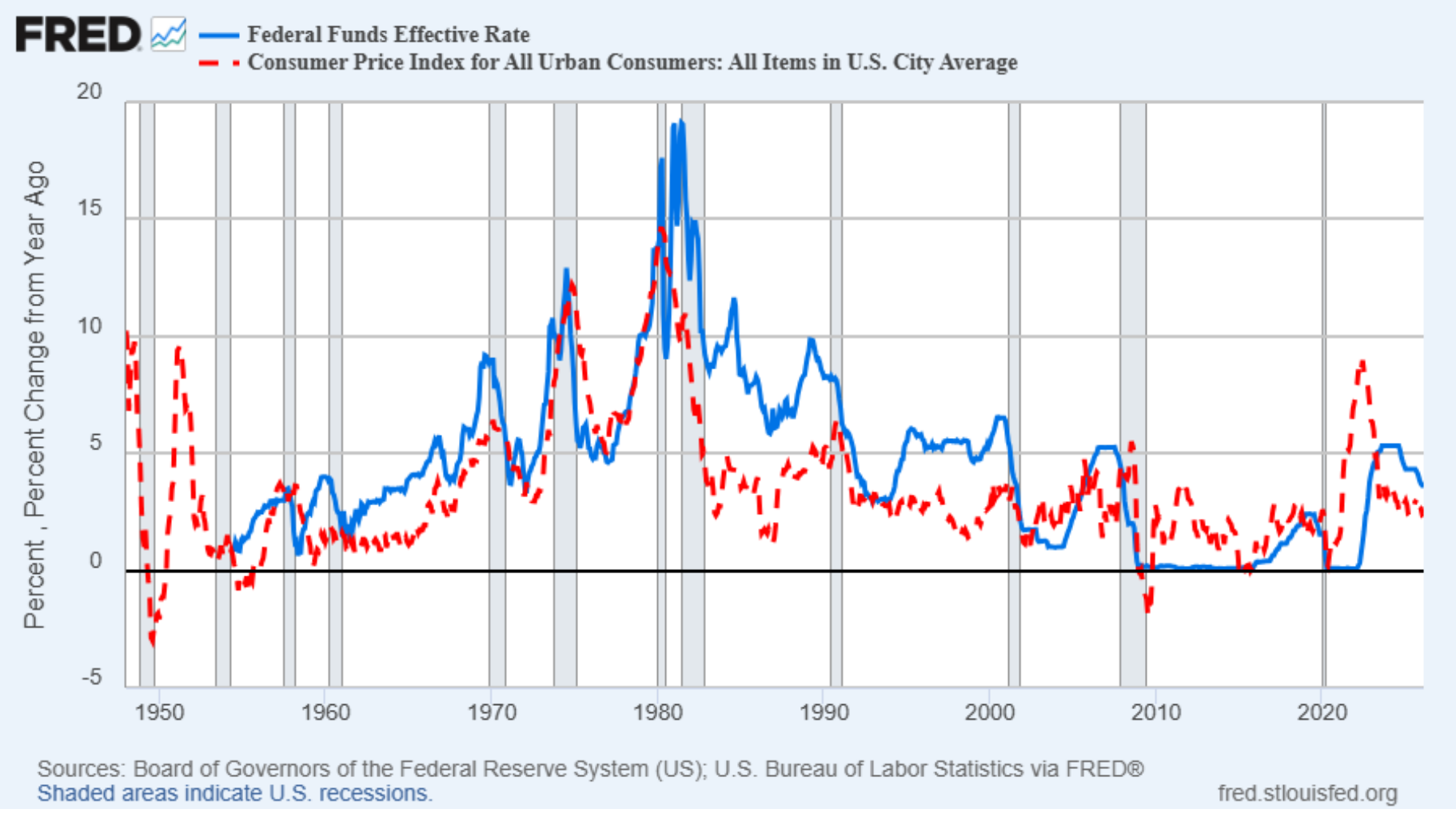
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Labor Force Participation Rate - 55 Yrs. & over
45.0
42.5
40.0
37.5
35.0
32.5
30.0
27.5
Percent
1950
1960
1970
1980
1990
2000
2010
2020
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
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FRED
Labor Force Participation Rate - 55 Yrs. & over
Highly persistent
Percent
45.0
42.5
40.0
37.5
35.0
32.5
30.0
27.5
1950
1960
1970
1980
1990
2000
2010
2020
Source: U.S. Bureau of Labor Statistics via FRED®
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Percent
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1970
1980
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2000
2010
2020
Break?
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
fred.stlouisfed.org



Causation vs. correlation: can the Federal Reserve control inflation?

Dearth of experiments. Policy-makers don't roll dice – need theory!

Forecasting

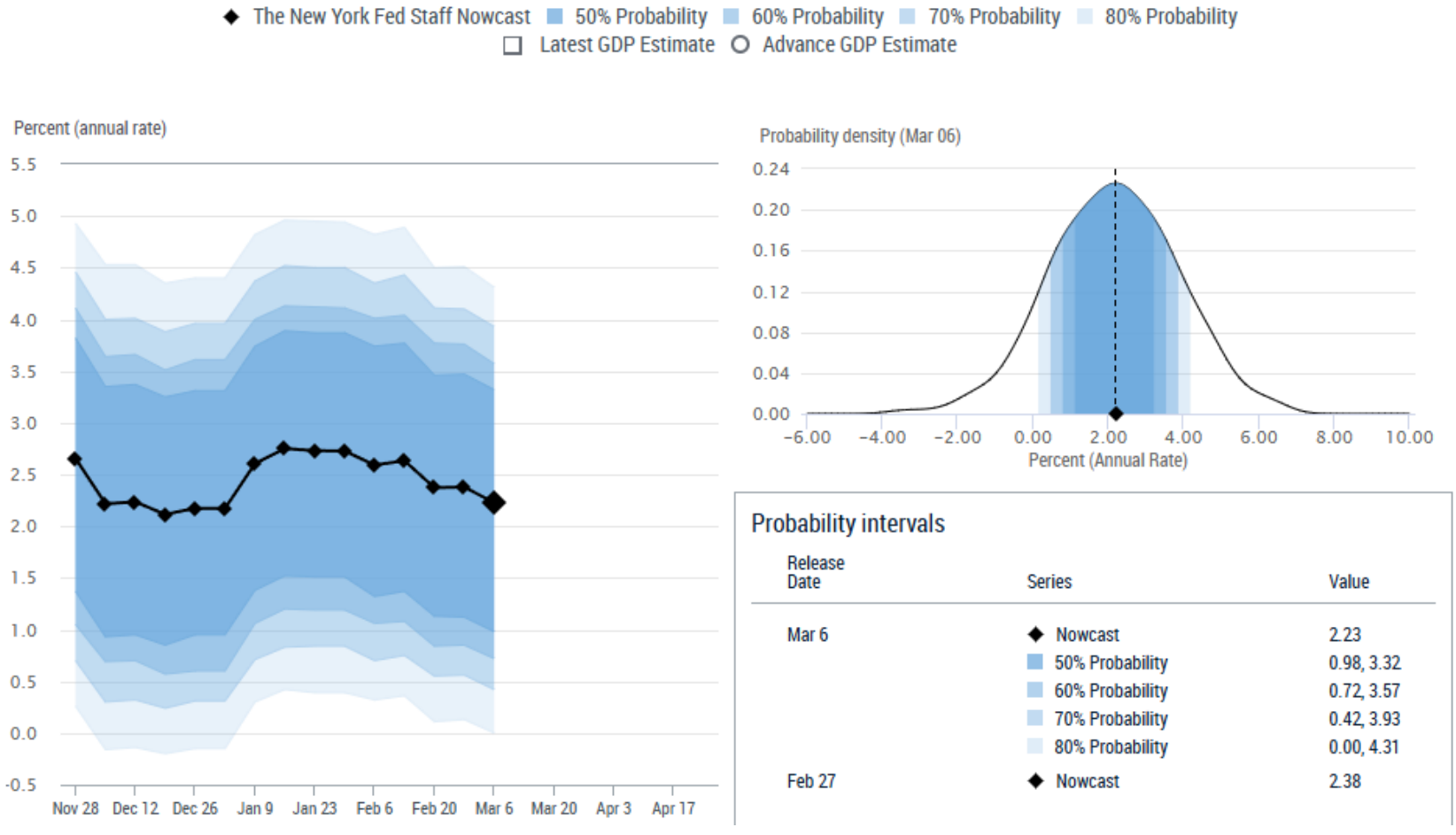
- The first part of this course will not consider **causal** questions: does X affect Y if we hold *everything else constant*?
  - We'll get back to these later – don't worry!
- Instead we will be concerned with **forecasting** (prediction): what is our best guess of the future value of Y, given what we observe up to the present?
- A variable might be helpful for forecasting without being causally important.
  - Example: the number of students carrying umbrellas to school in the morning is a good predictor of rainfall in the afternoon.
- Forecasting is a key role of economists in industry + government.

2026:Q1 | 2025:Q4 | 2025:Q3 | 2025:Q2

Latest Release: Mar 6, 2026

LAYOUT

New York Fed Staff Nowcast

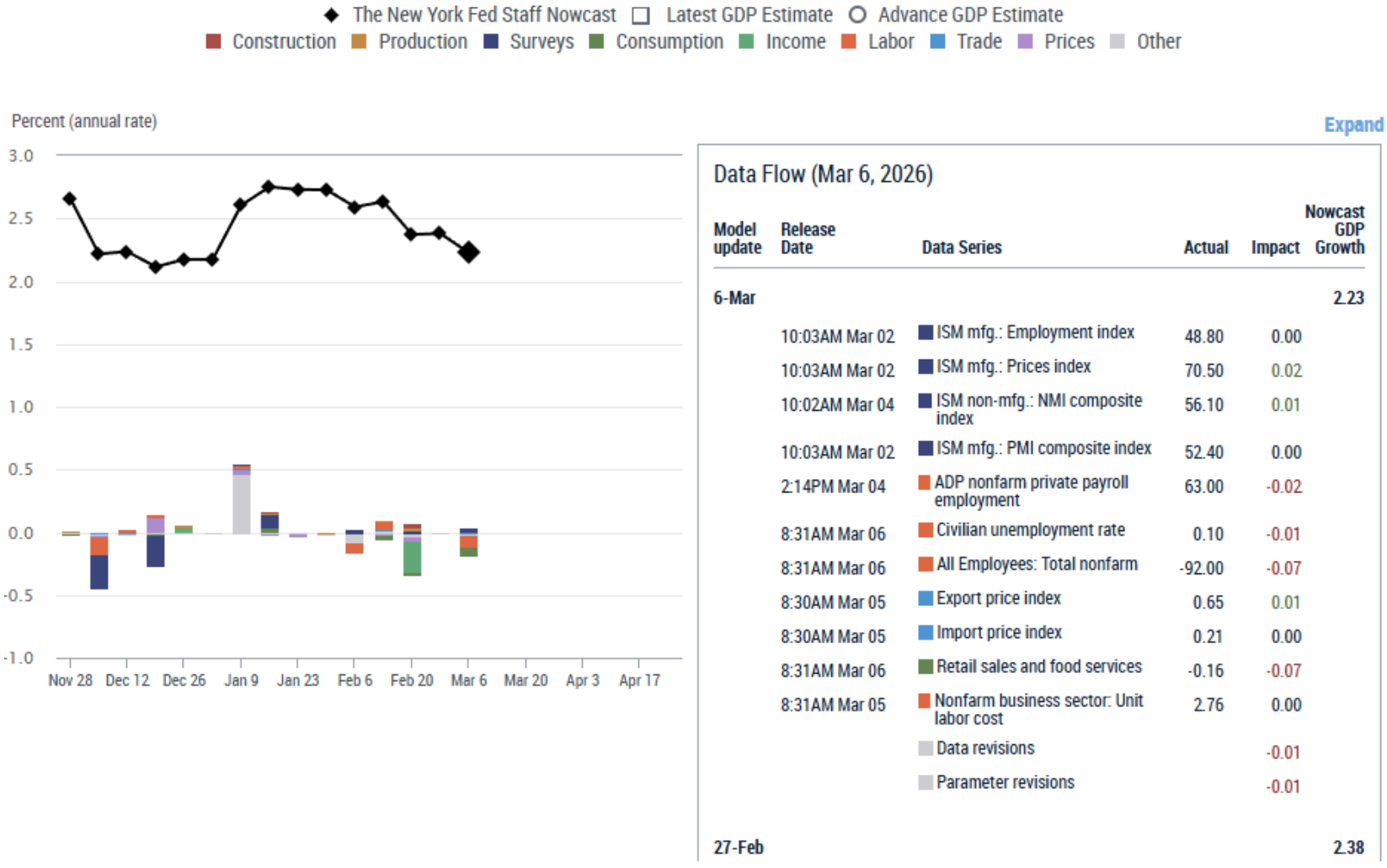


Probability intervals

Release Date	Series	Value
Mar 6	Nowcast	2.23
	50% Probability	0.98, 3.32
	60% Probability	0.72, 3.57
	70% Probability	0.42, 3.93
	80% Probability	0.00, 4.31
Feb 27	Nowcast	2.38

https://www.newyorkfed.org/research/policy/nowcast

Impact of Data Releases



Model update	Release Date	Data Series	Actual	Impact	Nowcast GDP Growth
6-Mar					2.23
	10:03AM Mar 02	ISM mfg.: Employment index	48.80	0.00	
	10:03AM Mar 02	ISM mfg.: Prices index	70.50	0.02	
	10:02AM Mar 04	ISM non-mfg.: NMI composite index	56.10	0.01	
	10:03AM Mar 02	ISM mfg.: PMI composite index	52.40	0.00	
	2:14PM Mar 04	ADP nonfarm private payroll employment	63.00	-0.02	
	8:31AM Mar 06	Civilian unemployment rate	0.10	-0.01	
	8:31AM Mar 06	All Employees: Total nonfarm	-92.00	-0.07	
	8:30AM Mar 05	Export price index	0.65	0.01	
	8:30AM Mar 05	Import price index	0.21	0.00	
	8:31AM Mar 06	Retail sales and food services	-0.16	-0.07	
	8:31AM Mar 05	Nonfarm business sector: Unit labor cost	2.76	0.00	
		Data revisions		-0.01	
		Parameter revisions		-0.01	
27-Feb					2.38

There's a wealth of public data series to use as predictors – but we will start by looking at small models and then expand from there.

Mathematical definition of a time series

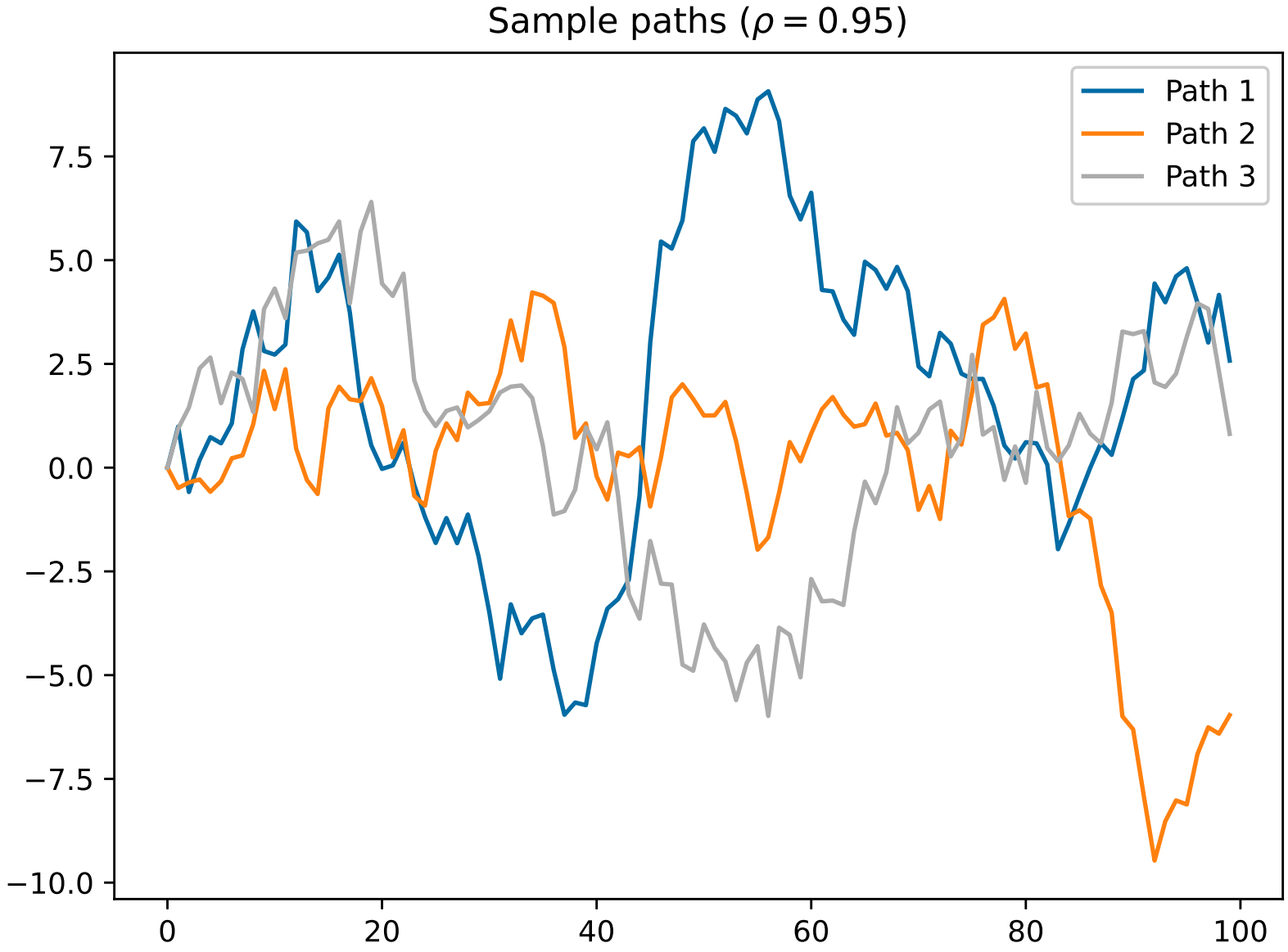
- A **time series** (or **stochastic process**) is a collection of random variables $y_1, y_2, y_3, \ldots$ whose ordering matters.
- We refer to the index t in y_t as "time".
  - Could be measured in days, months, quarters, years, etc.
- Being random variables, each y_t can be viewed as a function of an outcome ω in some latent sample space Ω: $y_t(\omega)$.
  - Think of Ω as all possible histories of the world, and ω as one realization.
- The time profile $y_1(\omega), y_2(\omega), y_3(\omega), \ldots$ is a **sample path** (or **realization**) of the time series. Conceptually, we can imagine alternative histories of the world, with different $\{y_t\}$ paths.
  - What if Napoleon had died from a stroke in 1802?
  - What if the Covid virus had mutated slightly differently?

Autoregressive process

- Consider a time series $\{y_t\}$ given by the recursive definition

$$y_t = \rho y_{t-1} + u_t, \quad \text{where} \quad u_t \overset{iid}{\sim} N(0,1) \quad \text{and} \quad y_0 = 0.$$

- This is called an **autoregression** (depends on its own past value).
- Think of u_t as random "**shocks**" that affect the evolution of the process (e.g., good/bad harvest, arrival of technological innovation).
  - To link with previous slide, u_t is implicitly itself a function of some underlying outcome ω, yielding the standard normal distribution.
- Next slide: three simulated sample paths for $t = 1, 2, \ldots, 100$ and $\rho = 0.95$.



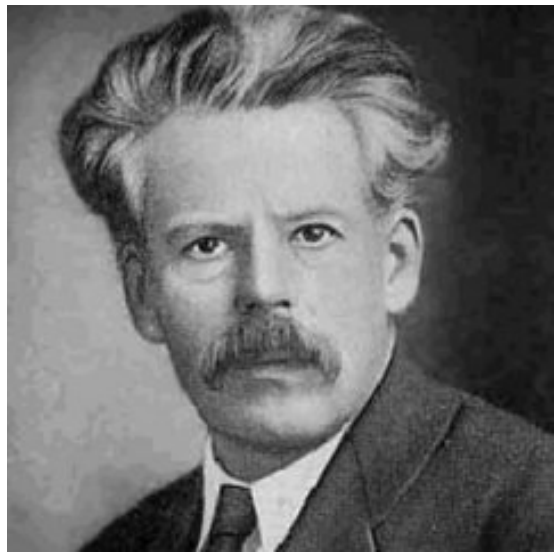
Each path is different, but they look deceptively like "typical" economic time series: persistent fluctuations, mixture of small+large movements

THE SUMMATION OF RANDOM CAUSES AS THE SOURCE OF CYCLIC PROCESSES*

By Eugen Slutzky

I. SCOPE OF THE INVESTIGATION

Almost all of the phenomena of economic life, like many other processes, social, meteorological, and others, occur in sequences of rising and falling movements, like waves. Just as waves following each other on the sea do not repeat each other perfectly, so economic cycles never repeat earlier ones exactly either in duration or in amplitude. Nevertheless, in both cases, it is almost always possible to detect, even in the multitude of individual peculiarities of the phenomena, marks of certain approximate uniformities and regularities. The eye of the



Impulse-propagation paradigm

- Slutsky (1927, previous slide), George Udny Yule (1927), and Ragnar Frisch (1933) developed the **impulse-propagation paradigm** for economic analysis.
- Impulse: random, external shocks that hit the economy.
- Propagation: the regular transmission mechanisms of the economy.
  - E.g., income falls $\rightarrow$ consumers cut back spending $\rightarrow$ businesses cut investment $\rightarrow \ldots$
- Hence, we get persistent cycles from irregular, unpredictable shocks.

> *If you hit a wooden rocking-horse with a club, the movement of the horse will be very different from that of the club.*
>
> — Ragnar Frisch, 1933

Stationarity

- If we want to forecast, we need to make some kind of assumption that "the future will be like the past".

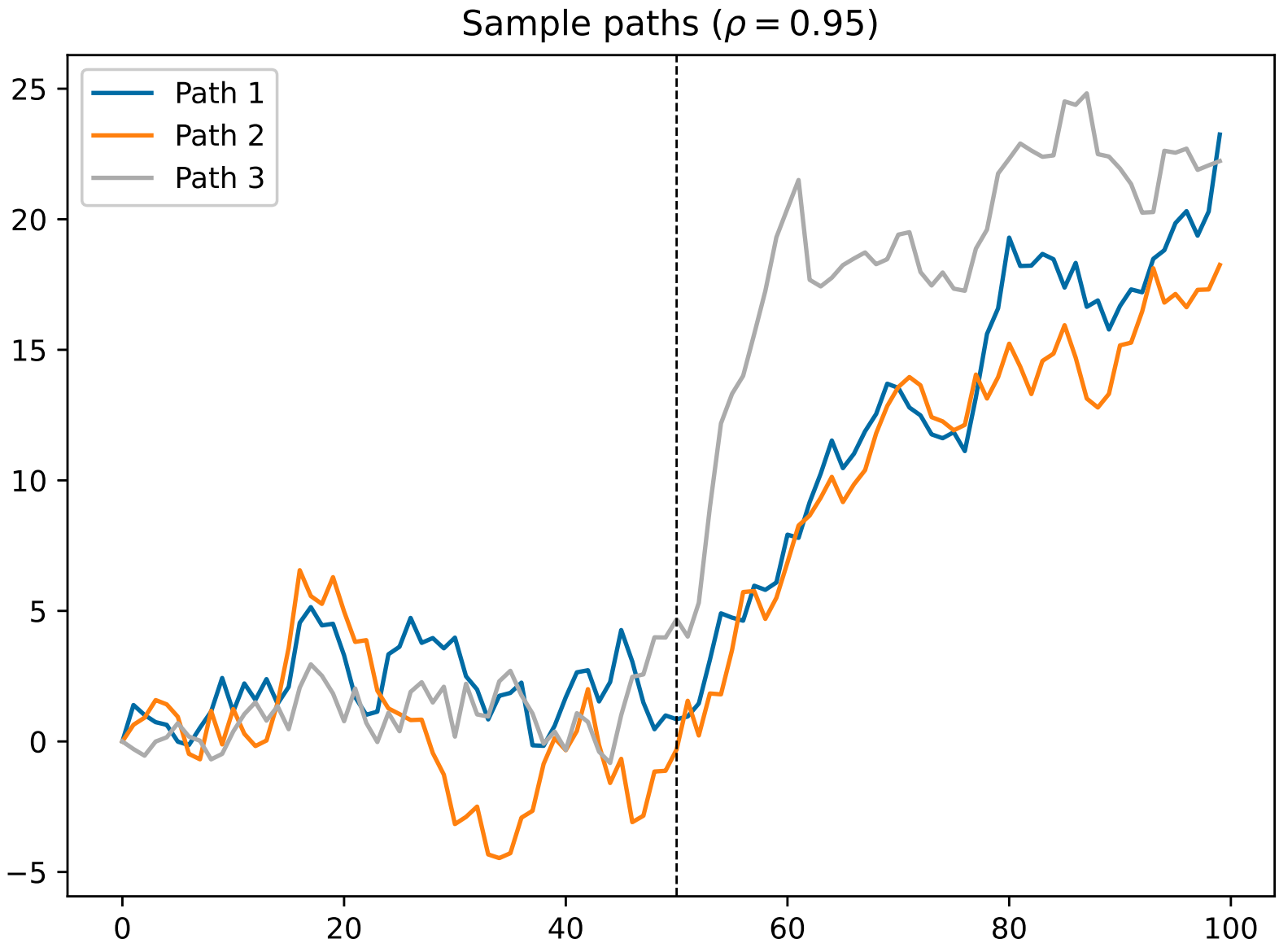
  That the sun will not rise tomorrow is no less intelligible a proposition, and implies no more contradiction than the affirmation, that it will rise. — David Hume, 1748

- The starting point is the following (will relax later).
- The time series $\{y_t\}$ is **(strictly) stationary** if for any integers k, ℓ, the distribution of $(y_t, y_{t+1}, \ldots, y_{t+k})$ is the same as that of $(y_{t+\ell}, y_{t+1+\ell}, \ldots, y_{t+k+\ell})$.
  - The distribution of (y_t, y_s) depends on t, s only through $t - s$.
- The *realized shocks* make y_t different from y_s for $s \neq t$, but *across histories of the world* each time point tends to behave the same (and similarly for each slice of consecutive time points).

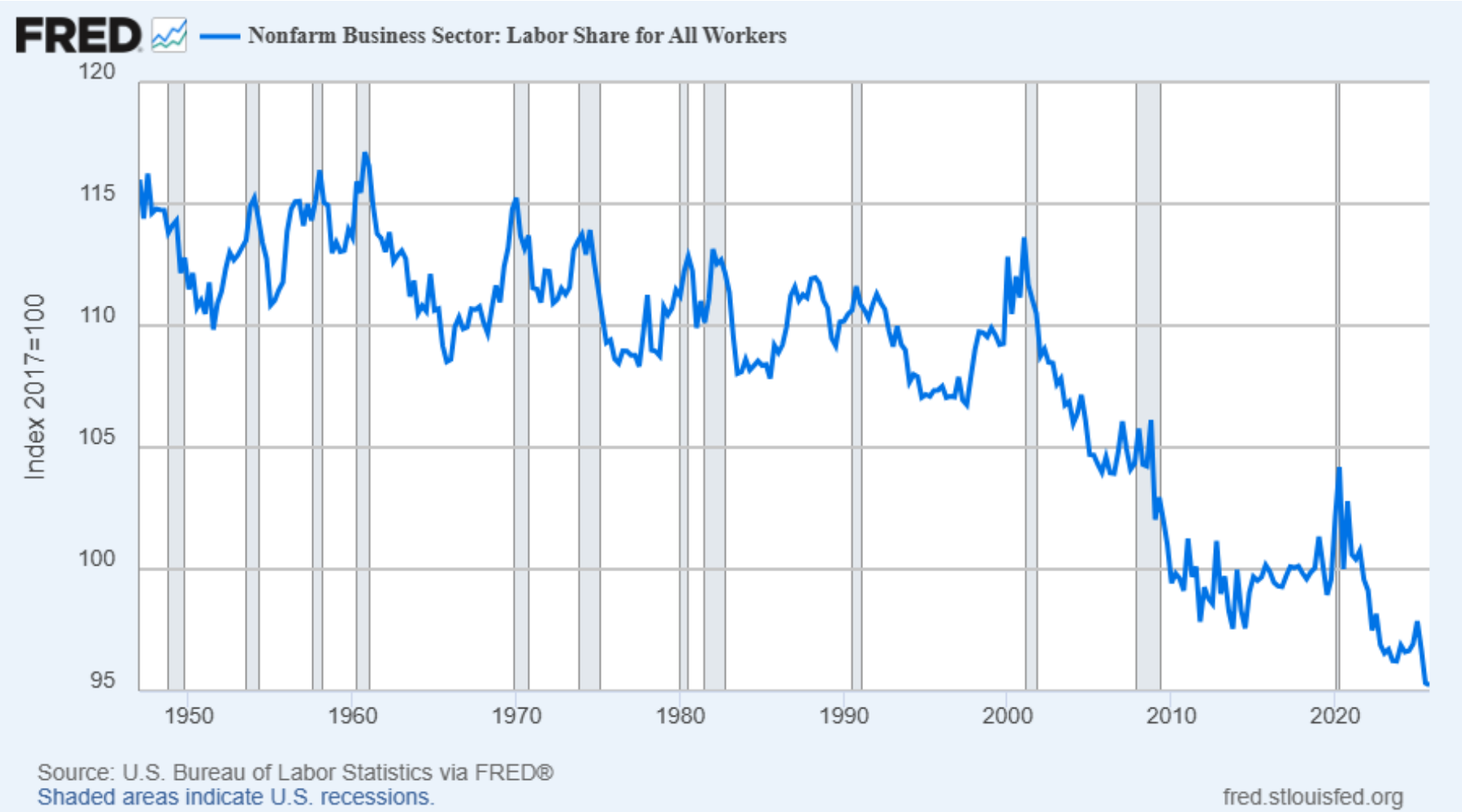
Non-stationarity

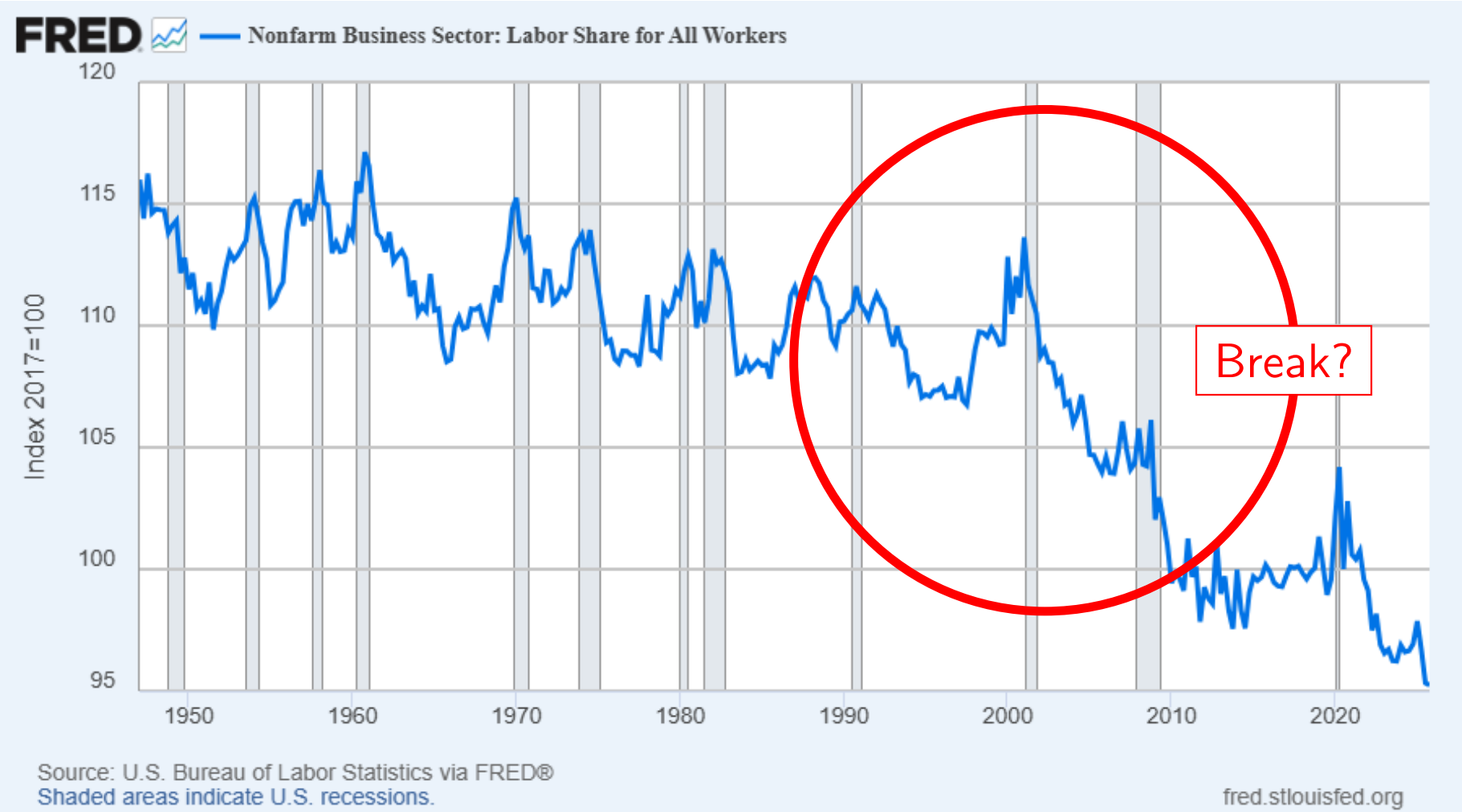
- We'll talk at length about non-stationary models later.
- Examples of non-stationarity:
  - A discrete break in the structure of the time series: distribution of y_t changes after the break.
  - A predictable trend in the series over time: average values are higher later in the sample than earlier.
  - Seasonality: e.g., January systematically different from February.
- Example of structural break:

$$y_t = c_t + \rho y_{t-1} + u_t, \quad \text{where} \quad c_t = \begin{cases} 0 & \text{for } t < t^*, \\ 1 & \text{for } t \geq t^*. \end{cases}$$



Break at time $t^* = 50$


FRED
Nonfarm Business Sector: Labor Share for All Workers
Index 2017=100
120
115
110
105
100
95
1950
1960
1970
1980
1990
2000
2010
2020
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
fred.stlouisfed.org


FRED
Nonfarm Business Sector: Labor Share for All Workers
Index 2017=100
120
115
110
105
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95
1950
1960
1970
1980
1990
2000
2010
2020
Break?
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
fred.stlouisfed.org

Differencing and logs

- If a time series does not look stationary, sometimes its **first differences** (changes) do:

$$\Delta y_t \equiv y_t - y_{t-1}.$$

- Many economic time series look reasonably stationary in *logged* first differences:

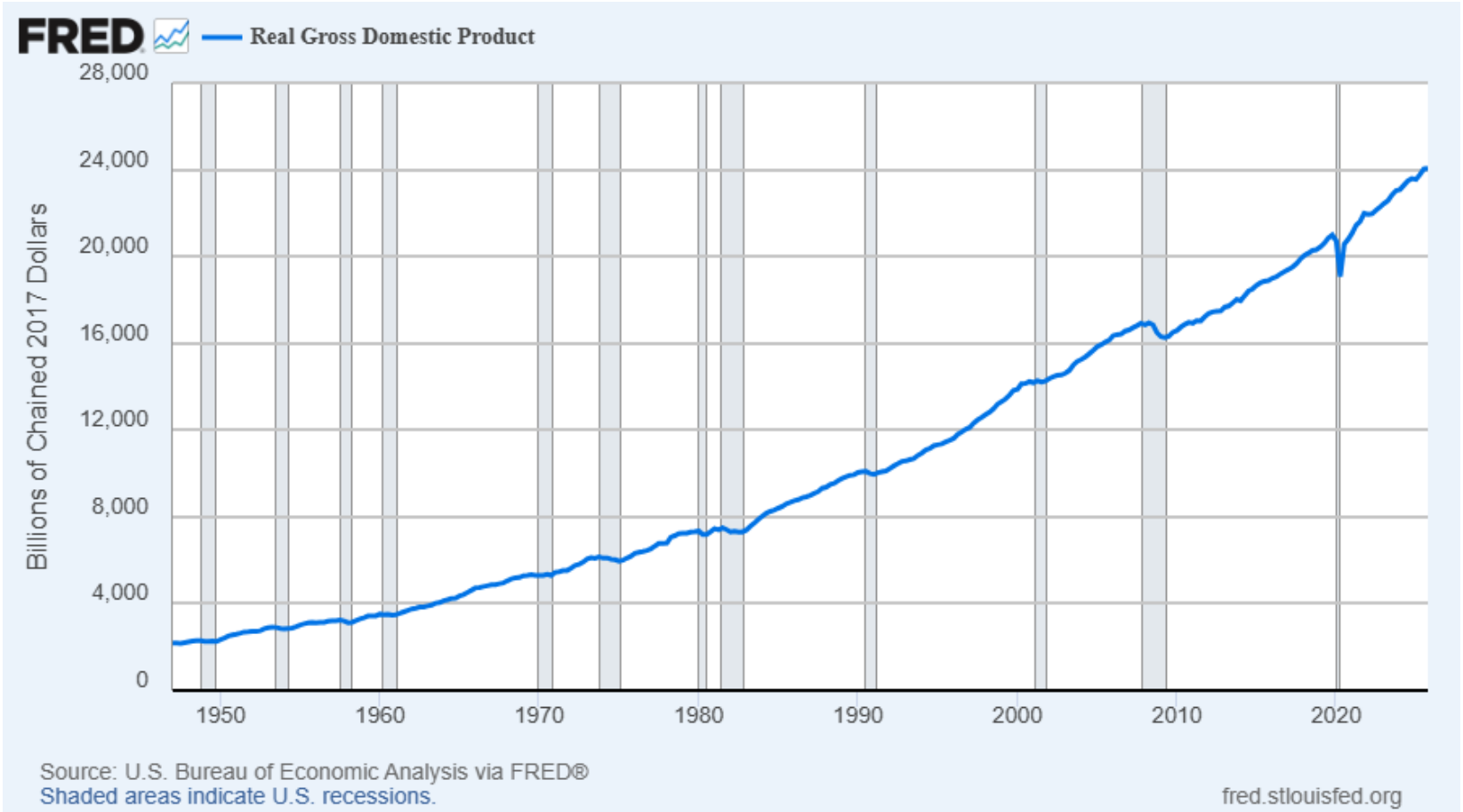
$$\Delta \log y_t \equiv \log y_t - \log y_{t-1}.$$

(In Econ, "log" always means natural logarithm.)

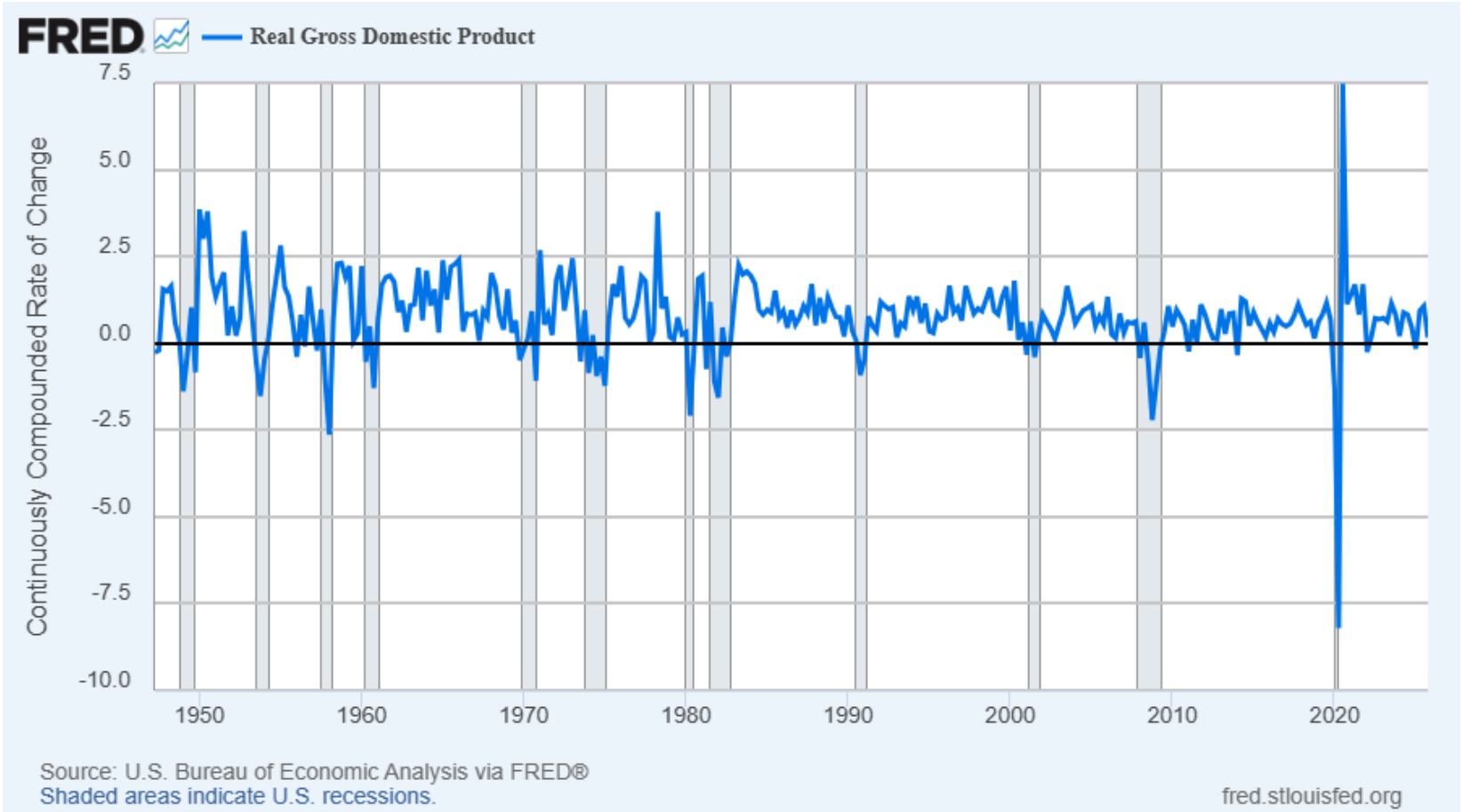
- This is because $\log(1+z) \approx z$ for small z (why?), so

$$\Delta \log y_t \approx \frac{y_t - y_{t-1}}{y_{t-1}} = \frac{1}{100} \times (\% \text{ change in } y_t),$$

and many economic series have fairly stable *growth rates*.



Real GDP in levels



Log growth rate of real GDP (quarterly)

Lags and leads

- Terminology:
  - y_{t-1} is the **lag** of y_t (prior observation).
  - $y_{t-\ell}$ is the ℓ-th lag of y_t (ℓ periods prior).
  - y_{t+1} is the **lead** of y_t (next observation).
  - $y_{t+\ell}$ is the ℓ-th lead of y_t (ℓ periods after).
- If we want to forecast, we need to know how y_t is related to its lags $y_{t-1}, y_{t-2}, \ldots$.

Autocorrelations

- Let's assume we have a stationary time series $\{y_t\}$.
- Then $E(y_t)$, $\text{Var}(y_t)$, and $\text{Cov}(y_t, y_{t-\ell})$ do not depend on t (for any integer ℓ).
- We call these the mean, variance, and **autocovariance** at **lag** ℓ.
- The function $\gamma(\ell) \equiv \text{Cov}(y_t, y_{t-\ell})$ is the **autocovariance function**.
  - Note: $\gamma(0) = \text{Var}(y_t)$.
- The function $\rho(\ell) \equiv \text{Corr}(y_t, y_{t-\ell}) = \frac{\gamma(\ell)}{\gamma(0)} \in [-1, 1]$ is the **autocorrelation function (ACF)**.
- The ACF satisfies $\rho(-\ell) = \rho(\ell)$.

Sample autocorrelation function

- Given data, we can estimate the autocorrelation in the obvious way:

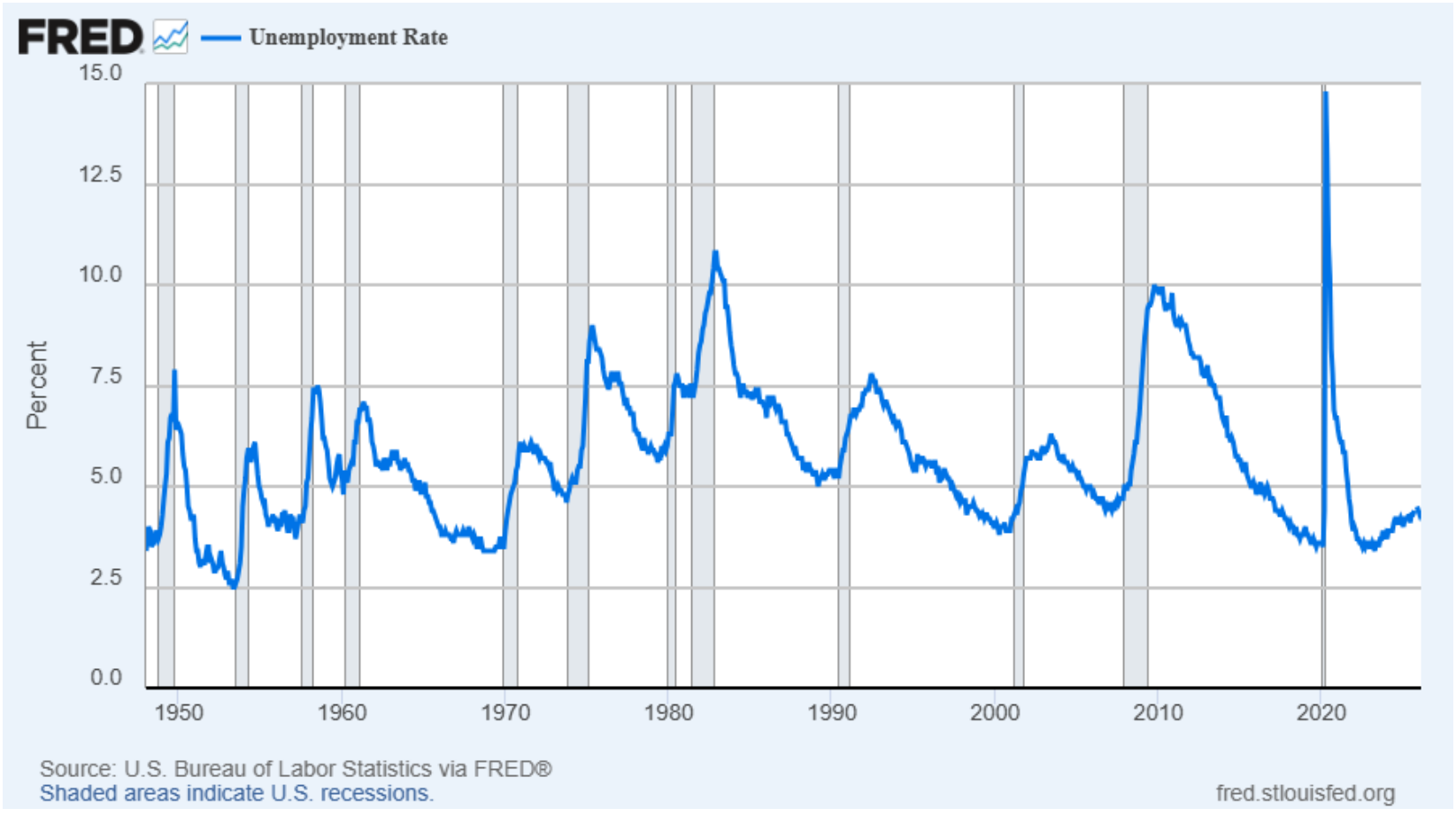
$$\hat{\rho}(\ell) \equiv \frac{\sum_{t=\ell+1}^{T}(y_t - \bar{y})(y_{t-\ell} - \bar{y})}{\sum_{t=1}^{T}(y_t - \bar{y})^2},$$

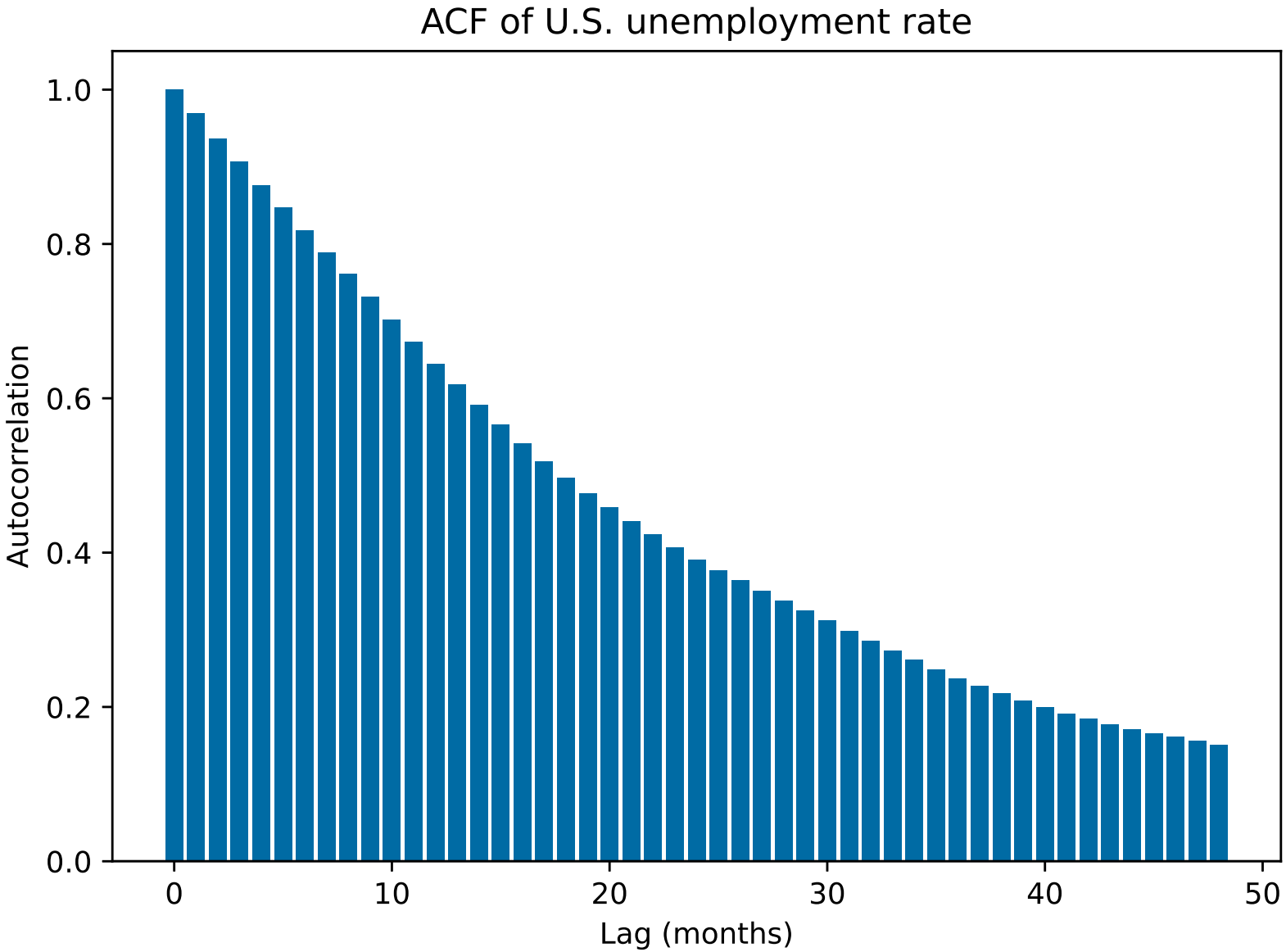
  where $\bar{y}$ is the sample average.

  - The function $\hat{\rho}(\ell)$, $\ell \geq 1$, is sometimes called the **correlogram**.

- Python:

```
from statsmodels.tsa.stattools import acf
rho = acf(y, nlags=48)
```


FRED
Unemployment Rate
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Percent
1950
1960
1970
1980
1990
2000
2010
2020
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
fred.stlouisfed.org



$\hat{\rho}(\ell) > 0$ for $\ell = 1, 2, \ldots$: the series is **positively serially correlated**.

The persistence is quite strong. $\hat{\rho}(\ell) > 0.2$ even after 3 years.

Covariance stationarity

- We will often only be concerned with the first and second moments of time series (mean, autocovariances).
- It then makes sense to work with the following weaker stationarity concept.
- $\{y_t\}$ is **covariance stationary** if neither its mean nor its autocovariances depend on time.

$$\text{strict stationarity} + \text{variance exists} \implies \text{covariance stationarity}$$

$$\text{covariance stationarity} \not\implies \text{strict stationarity}$$

White noise

- As a building block for more complicated dynamic models (next lectures), it's useful to define a process that is *not* serially correlated.
- A covariance stationary series $\{y_t\}$ is **white noise** if:
  1. $E(y_t) = 0$.
  2. $\gamma(\ell) = 0$ for $\ell \geq 1$ (**serially uncorrelated**).
- For a white noise process, there's no correlation between the *level* of the series at time t and the level at any other time s.
  - But there could be serial correlation in $|y_t|$, say.
- Special case of white noise process: martingale difference sequence (next slide).

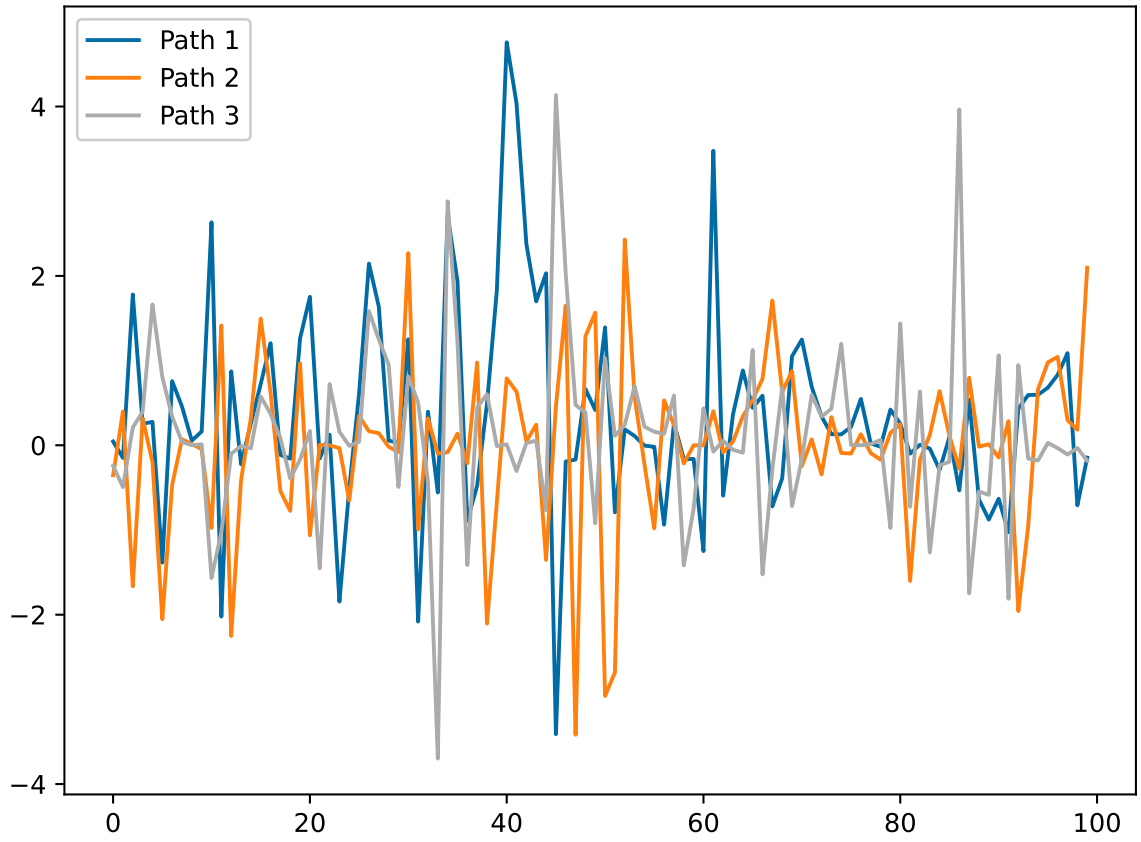
Martingale difference sequence

- A series $\{y_t\}$ is called a **martingale** with respect to the **filtration** (sequence of information sets) $\{\mathcal{F}_t\}_t$ if:
  1. $y_t \in \mathcal{F}_t$ for all t (**adapted** to the filtration).
  2. $E[y_t \mid \mathcal{F}_{t-1}] = y_{t-1}$ (best forecast = no change).
- Typically, $\mathcal{F}_t = \sigma(y_t, y_{t-1}, y_{t-2}, \dots)$, the information generated by past data.
- A series $\{y_t\}$ is called a **martingale difference sequence** (MDS) if it's adapted to $\mathcal{F}_t$ and

$$E[y_t \mid \mathcal{F}_{t-1}] = 0.$$

  - Real-world example: stock returns (efficient markets).
- If $\{y_t\}$ is a martingale, then Δy_t is a MDS.

Martingale difference sequence: example



$$y_t = |u_{t-1}|u_t, \quad u_t \overset{iid}{\sim} N(0,1)$$

Exercise: show that $\{y_t\}$ is a MDS. But it's not independent over time: $\text{Cov}(\log|y_t|, \log|y_{t-1}|) = \text{Var}(\log|u_t|) \neq 0$.

Relationship between concepts

- Exercise: prove that for a time series $\{y_t\}$ with mean zero and finite variance that is constant through time,

$$\text{i.i.d.} \quad \Longrightarrow \quad \text{MDS} \quad \Longrightarrow \quad \text{white noise.}$$

- Here the MDS is wrt. the natural filtration $\mathcal{F}_t = \sigma(y_t, y_{t-1}, y_{t-2}, \dots)$.

What did we learn?

- Mathematical definition of time series
- Example: AR process
- Stationarity, differencing
- Concepts related to serial correlation
  - ACF: population, sample
  - White noise, martingale difference sequence

Looking ahead

- How to model time series dependence?
- Autoregressive model
  - Stationarity
  - Moments
  - Forecasting
  - Choice of lag length